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UMN
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NUSANTARA

IF351 DATABASE SYSTEMS

02 FUNCTIONAL DEPENDENCIES AND NORMALIZATION FOR RELATIONAL DATABASES



REVIEW: DATABASE SECURITY

- Introduction to Database Security Issues
- Concept of a View
- Discretionary Access Control
- Mandatory Access Control
- Adding & Removing Users
- Granting, Passing On, & Revoking Privileges
- Creating, Updating, & Deleting Views

COURSE SUB LEARNING OUTCOMES (SUB-CLO)

- Sub-CLO 12
 - Students are able to determine functional dependencies in unnormalized relations (C3)
- Sub-CLO 13
 - Students are able to apply database normalization technique (C3)

OUTLINE

- Functional Dependencies
- First Normal Form (1NF)
- Partial Dependency
- Second Normal Form (2NF)
- Transitive Dependency
- Third Normal Form (3NF)

FUNCTIONAL DEPENDENCIES

- A functional dependency is a constraint between two sets of attributes from the database
- $X \rightarrow Y$
 - Y depends on X
 - Y is determined by X
 - The values of X determine the values of Y
 - Y is functionally dependent on X
- The set of attributes X is called the left-hand side of the functional dependency
- Y is called the right-hand side
- X functionally determines Y in a relation schema R if, and only if, whenever two tuples of $r(R)$ agree on their X -value, they must necessarily agree on their Y -value

FUNCTIONAL DEPENDENCIES

- A functional dependency is a property of the semantics or meaning of the attributes
 - Employee(EmployeeID, EmployeeName)
 - EmployeeID $\rightarrow$ EmployeeName
 - Project(ProjectNumber, ProjectName, ProjectLocation)
 - ProjectNumber $\rightarrow$ { ProjectName, ProjectLocation }

NORMALIZATION OF RELATIONS

- The normalization process, as first proposed by Codd (1972), takes a relation schema through a series of tests to certify whether it satisfies a certain normal form
- Unsatisfactory relation schemas that do not meet certain conditions are decomposed into smaller relation schemas that meet the tests and hence possess the desirable properties

NORMALIZATION OF RELATIONS

Objectives

- Minimizing redundancy
- Minimizing the insertion, deletion, and update anomalies

Normal Forms

- First Normal Form (1NF)
- Second Normal Form (2NF)
- Third Normal Form (3NF)
- Boyce-Codd Normal Form (BCNF)
- Fourth Normal Form (4NF)
- Fifth Normal Form (5NF)
- Domain-Key Normal Form (DKNF)

FIRST NORMAL FORM (1NF)

- First normal form is defined to disallow **multivalued attributes**, **composite attributes**, and their combinations
- The domain of an attribute must include only **atomic** (simple, indivisible) values and that the value of any attribute in a tuple must be a **single value** from the domain of that attribute

FIRST NORMAL FORM (1NF)

- A relation schema that is not in 1NF

DEPARTMENT

Dname	<u>Dnumber</u>	Dmgr_ssn	Dlocations

- Sample state of relation Department

DEPARTMENT

Dname	<u>Dnumber</u>	Dmgr_ssn	Dlocations
Research	5	333445555	{Bellaire, Sugarland, Houston}
Administration	4	987654321	{Stafford}
Headquarters	1	888665555	{Houston}

- 1NF version of relation Department

DEPARTMENT

Dname	<u>Dnumber</u>	Dmgr_ssn	<u>Dlocation</u>
Research	5	333445555	Bellaire
Research	5	333445555	Sugarland
Research	5	333445555	Houston
Administration	4	987654321	Stafford
Headquarters	1	888665555	Houston

FIRST NORMAL FORM (1NF)

Student	SSN	SName	Major	GPA
	111223333	Smith, Tom	History	3.1
	222334444	Jones, Mary	Math	3.1
	333445555	Lee, Pamela	CSC Math	3.8
	444556666	Burns, Edward	MIS Finance	3.2
	555667777	Jones, Rose	MIS	3.3

SSN is the primary key of the Student relation.



1NF Normalization

Student1	SSN	SName	Major	GPA
	111223333	Smith, Tom	History	3.1
	222334444	Jones, Mary	Math	3.1
	333445555	Lee, Pamela	CSC	3.8
	333445555	Lee, Pamela	Math	3.8
	444556666	Burns, Edward	MIS	3.2
	444556666	Burns, Edward	Finance	3.2
	555667777	Jones, Rose	MIS	3.3

Now, SSN and Major together become the primary key of the Student1 relation.

FULL FUNCTIONAL DEPENDENCY VS PARTIAL DEPENDENCY

Full Functional Dependency

- A functional dependency $X \rightarrow Y$ is a full functional dependency if Y is functionally dependent on X but **not functionally dependent on any proper subset of X**
- A functional dependency $X \rightarrow Y$ is a full functional dependency if removal of any attribute A from X means that the dependency does not hold any more
 - For any attribute $A \in X$, $(X - \{A\})$ does not functionally determine Y

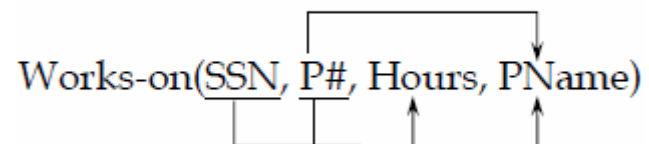
Partial Dependency

- A functional dependency $X \rightarrow Y$ is a partial dependency if Y is functionally dependent on X and **also functionally dependent on any proper subset of X**
- A functional dependency $X \rightarrow Y$ is a partial dependency if some attribute $A \in X$ can be removed from X and the dependency still holds
 - For some $A \in X$, $(X - \{A\}) \rightarrow Y$

FULL FUNCTIONAL DEPENDENCY VS PARTIAL DEPENDENCY

- $\text{WorksOn}(\text{EmployeeID}, \text{ProjectNumber}, \text{Hours}, \text{ProjectName})$
 - $\text{EmployeeID}, \text{ProjectNumber} \rightarrow \text{Hours}, \text{ProjectName}$
 - $\text{ProjectNumber} \rightarrow \text{ProjectName}$
- $\text{EmployeeID}, \text{ProjectNumber} \rightarrow \text{Hours}$ is a full functional dependency
 - Neither $\text{EmployeeID} \rightarrow \text{Hours}$ nor $\text{ProjectNumber} \rightarrow \text{Hours}$ holds
- $\text{EmployeeID}, \text{ProjectNumber} \rightarrow \text{ProjectName}$ is a partial dependency
 - $\text{ProjectNumber} \rightarrow \text{ProjectName}$ holds

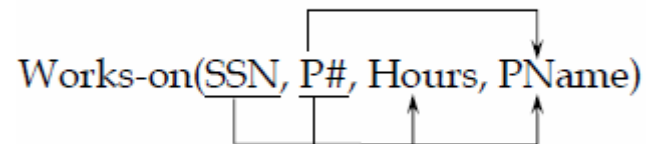
ANOMALY RESULTED FROM PARTIAL DEPENDENCY



SSN	P#	Hours	PName
123456789	3	15	Multimedia
333445555	3	10	Multimedia
888665555	3	8	Multimedia
987654321	1	10	Reorganization
888665555	1	20	Reorganization
555334444	6	25	Company DB

What's wrong with this table?
Any problem?

ANOMALY RESULTED FROM PARTIAL DEPENDENCY

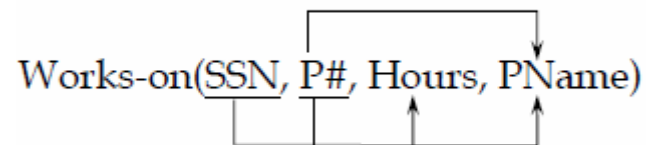


SSN	P#	Hours	PName
123456789	3	15	Multimedia
333445555	3	10	Multimedia
888665555	3	8	Multimedia
987654321	1	10	Reorganization
888665555	1	20	Reorganization
555334444	6	25	Company DB

Insertion Anomaly

- There is no way to insert a new project that has no employee assigned to the project
- For example, P# is 7 and PName is Algorithm

ANOMALY RESULTED FROM PARTIAL DEPENDENCY

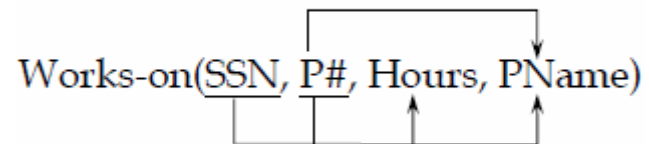


SSN	P#	Hours	PName
123456789	3	15	Multimedia
333445555	3	10	Multimedia
888665555	3	8	Multimedia
987654321	1	10	Reorganization
888665555	1	20	Reorganization
555334444	6	25	Company DB

Deletion Anomaly

- If a tuple that represents the only employee working on a particular project is deleted, the information about that project is lost from the database
- For example, if a tuple with SSN 555334444 is deleted

ANOMALY RESULTED FROM PARTIAL DEPENDENCY

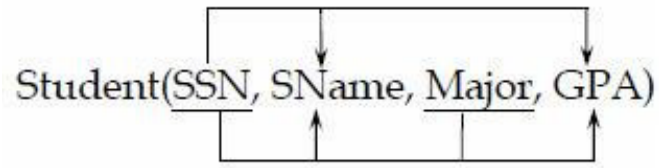


SSN	P#	Hours	PName
123456789	3	15	Multimedia
333445555	3	10	Multimedia
888665555	3	8	Multimedia
987654321	1	10	Reorganization
888665555	1	20	Reorganization
555334444	6	25	Company DB

Update Anomaly

- If PName of a particular project is changed, PName of the same P# in other tuples must also be updated
- Otherwise, the database will become inconsistent
- For example, PName of P# 3 is changed

ANOMALY RESULTED FROM PARTIAL DEPENDENCY



SSN	SName	Major	GPA
111223333	Smith, Tom	History	3.1
222334444	Jones, Mary	Math	3.1
333445555	Lee, Pamela	CSC	3.8
333445555	Lee, Pamela	Math	3.8
444556666	Burns, Edward	MIS	3.2
444556666	Burns, Edward	Finance	3.2
555667777	Jones, Rose	MIS	3.3

Insertion Anomaly?

Deletion Anomaly?

Update Anomaly?

SECOND NORMAL FORM (2NF)

A relation schema R is in 2NF if
it is in 1NF
and
every nonprime attribute A in R
is fully functionally dependent on
the primary key of R

Second normal form aims at eliminating those modification anomalies caused by the presence of **partial functional dependencies** whose left-hand side attributes are part of the primary key of a relation

SECOND NORMAL FORM (2NF)

- If the primary key contains a single attribute, the test need not be applied at all
- If a relation is in 1NF and its key consists of a single attribute, the relation is automatically in 2NF
- The only time we have to concern about 2NF is when the key of the relation is composite

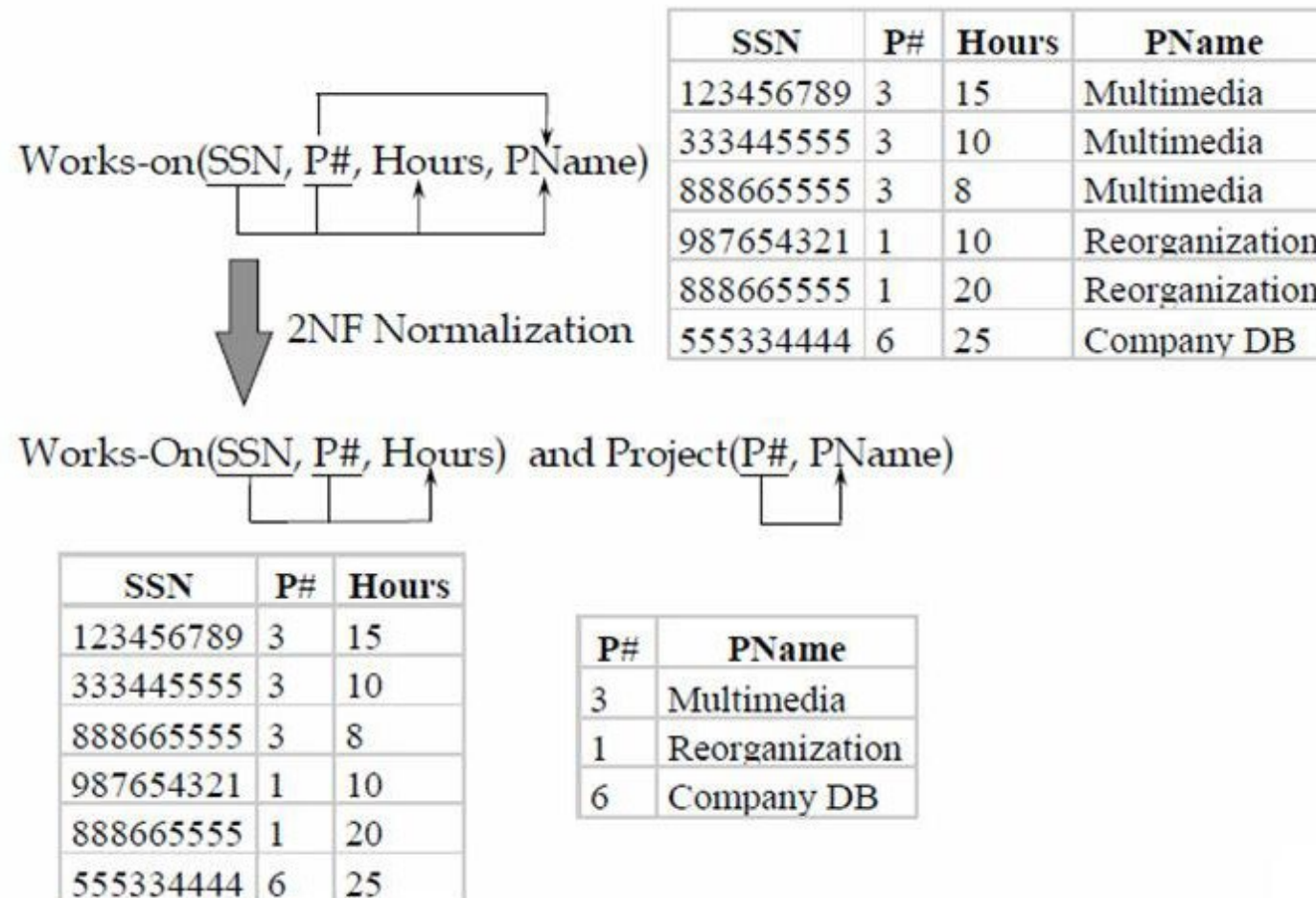
SECOND NORMAL FORM (2NF)

If a relation schema is not in 2NF, it can be second normalized or 2NF normalized into a number of 2NF relations

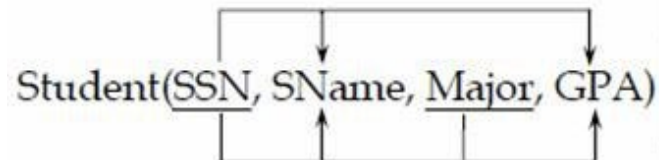
2NF Normalization

- Decompose and set up a new relation for each partial key with its dependent attribute(s)
- Make sure to keep a relation with the original primary key and any attributes that are fully functionally dependent on it

SECOND NORMAL FORM (2NF)



SECOND NORMAL FORM (2NF)



2NF Normalization

Student(SSN, SName, GPA) and Student-Major(SSN, Major)

SSN	SName	GPA
111223333	Smith, Tom	3.1
222334444	Jones, Mary	3.1
333445555	Lee, Pamela	3.8
444556666	Burns, Edward	3.2
555667777	Jones, Rose	3.3

SSN	SName	Major	GPA
111223333	Smith, Tom	History	3.1
222334444	Jones, Mary	Math	3.1
333445555	Lee, Pamela	CSC	3.8
333445555	Lee, Pamela	Math	3.8
444556666	Burns, Edward	MIS	3.2
444556666	Burns, Edward	Finance	3.2
555667777	Jones, Rose	MIS	3.3

SSN	Major
111223333	History
222334444	Math
333445555	CSC
333445555	Math
444556666	MIS
444556666	Finance
555667777	MIS

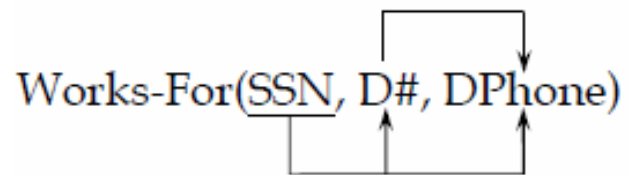
TRANSITIVE DEPENDENCY

- A functional dependency $X \rightarrow Y$ in a relation schema R is a transitive dependency if there is a set of attributes Z that is neither a candidate key nor a subset of any key of R , and both $X \rightarrow Z$ and $Z \rightarrow Y$ hold

TRANSITIVE DEPENDENCY

- Works-For(SSN, D#, DPhone)
 - $SSN \rightarrow D\#, DPhone$
 - $D\# \rightarrow DPhone$
- The dependency $SSN \rightarrow DPhone$ is transitive through D#
 - Both the dependencies $SSN \rightarrow D\#$ and $D\# \rightarrow DPhone$ hold
 - D# is neither a key itself nor a subset of the key of Works-For
- $SSN \rightarrow D\#$ is not a transitive dependency

ANOMALY RESULTED FROM TRANSITIVE DEPENDENCY



SSN	D#	DPhone
123456789	5	621-2144
333445555	5	621-2144
888665555	1	621-1000
987654321	4	621-1500
666112222	6	621-2200
555334444	6	621-2200

Insertion Anomaly?

Deletion Anomaly?

Update Anomaly?

THIRD NORMAL FORM (3NF)

A relation schema R is in 3NF if
it satisfies 2NF
and
no nonprime attribute of R
is transitively dependent on the primary key

Third normal form aims at eliminating those modification anomalies caused by the presence of **transitive functional dependencies**

THIRD NORMAL FORM (3NF)

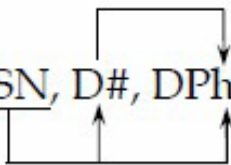
If a relation is not in 3NF, it can be normalized into a number of 3NF relations

3NF Normalization

- Decompose and set up a relation that includes the nonkey attribute(s) that functionally determine(s) other nonkey attribute(s)
- The determinant becomes the primary key of the new relation

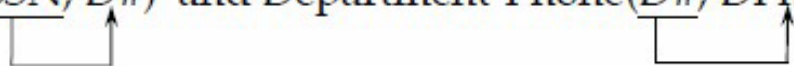
THIRD NORMAL FORM (3NF)

Works-For(SSN, D#, DPhone)



3NF Normalization

Works-For(SSN, D#) and Department-Phone(D#, DPhone)



SSN	D#	DPhone
123456789	5	621-2144
333445555	5	621-2144
888665555	1	621-1000
987654321	4	621-1500
666112222	6	621-2200
555334444	6	621-2200

SSN	D#
123456789	5
333445555	5
888665555	1
987654321	4
666112222	6
555334444	6

D#	DPhone
5	621-2144
1	621-1000
4	621-1500
6	621-2200

PRACTICAL CONSIDERATIONS

- Higher normal forms eliminate more modification anomalies than lower normal forms do
 - Therefore, the general idea of normalization: the database designer should aim for relations in the highest normal form
- However, higher normal forms impose more overhead on retrieval requests because when moving towards higher normal forms, a relation may be decomposed into several smaller relations
 - Join operations which are not required by the initial relation may be required by these smaller relations
- In practice, a relation in 3NF or BCNF will be considered as a “good” relation

EXAMPLE

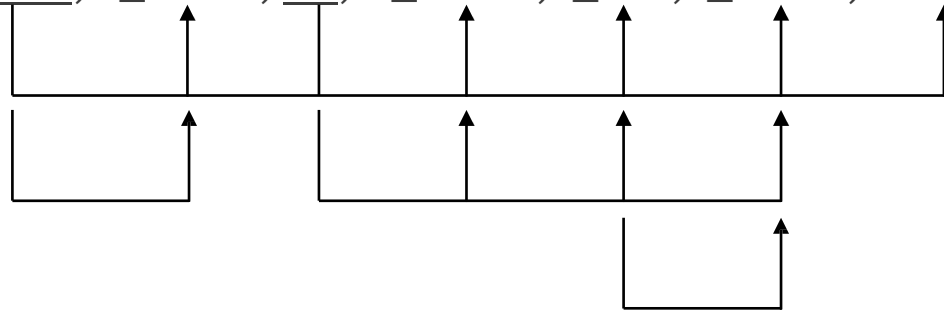
- CourseTaking (S_NIM, S_Name, C#, C_Name, I_NIK, I_Name, Grade)
 - S_NIM and S_Name represent the student number and name
 - C# and C_Name represent the course number and name
 - I_NIK and I_Name represent the instructor number and name
 - Grade is the grade of a particular student for a particular course
 - Since a student can take more than one course at one time and a course can only be taught by one instructor, S_NIM and C# are sufficient enough to be the primary key of the CourseTaking relation
 - S_NIM uniquely determines S_Name
 - C# uniquely determines C_Name, I_NIK, I_Name
 - I_NIK uniquely determines I_Name

EXAMPLE

- CourseTaking (S_NIM, S_Name, C#, C_Name, I_NIK, I_Name, Grade)
 - $S_NIM, C\# \rightarrow S_Name, C_Name, I_NIK, I_Name, Grade$
 - $S_NIM \rightarrow S_Name$
 - $C\# \rightarrow C_Name, I_NIK, I_Name$
 - $I_NIK \rightarrow I_Name$

EXAMPLE

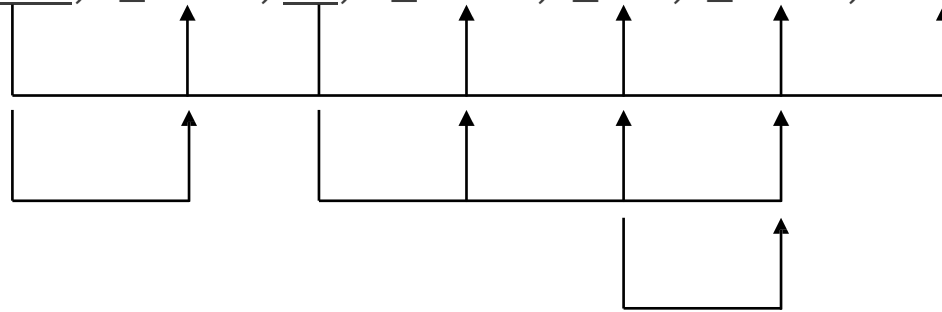
- CourseTaking (S_NIM, S_Name, C#, C_Name, I_NIK, I_Name, Grade)



- CourseTaking is in 1NF since every attribute is single-valued for each tuple in the relation

EXAMPLE

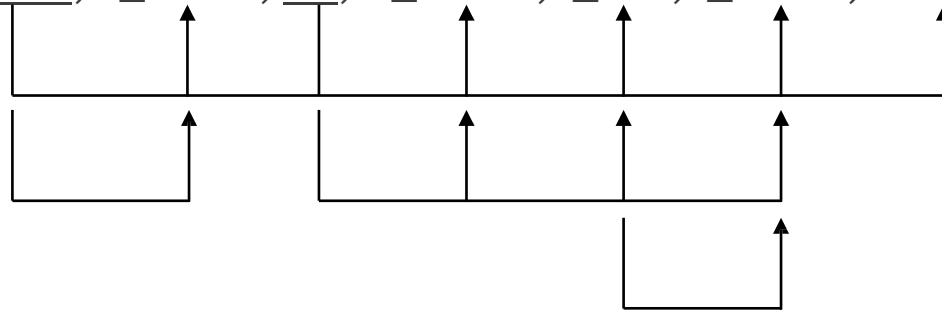
- CourseTaking (S_NIM, S_Name, C#, C_Name, I_NIK, I_Name, Grade)



- Partial Dependencies
 - $S_NIM, C\# \rightarrow S_Name$ since $S_NIM \rightarrow S_Name$ holds
 - $S_NIM, C\# \rightarrow C_Name, I_NIK, I_Name$ since $C\# \rightarrow C_Name, I_NIK, I_Name$ holds

EXAMPLE

- CourseTaking (S_NIM, S_Name, C#, C_Name, I_NIK, I_Name, Grade)



- 2NF Normalization
 - Student (S_NIM, S_Name)
 - CourseTeaching (C#, C_Name, I_NIK, I_Name)
 - CourseTaking (S_NIM, C#, Grade)

EXAMPLE

In 2NF and also 3NF

- Student (S_NIM, S_Name)

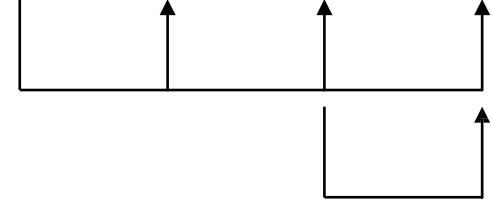


- CourseTaking (S_NIM, C#, Grade)



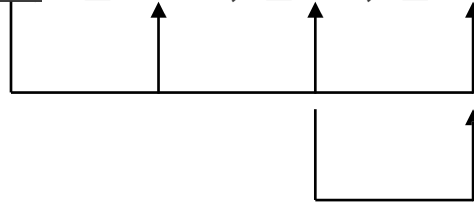
In 2NF but not in 3NF

- CourseTeaching (C#, C_Name, I_NIK, I_Name)



EXAMPLE

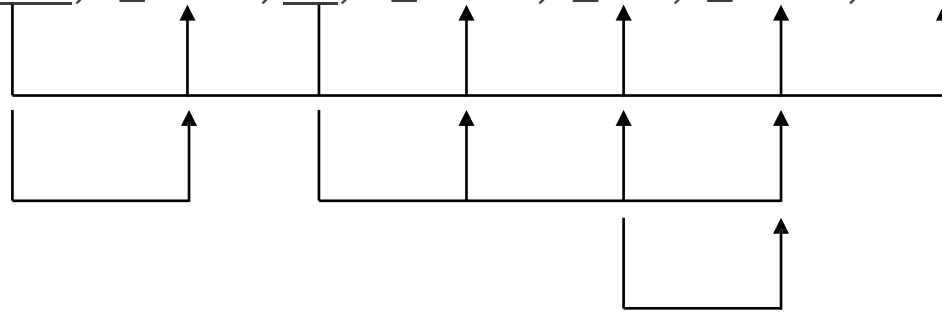
- CourseTeaching (C#, C_Name, I_NIK, I_Name)



- Transitive Dependency
 - $C\# \rightarrow I_Name$ since $C\# \rightarrow I_NIK$ and $I_NIK \rightarrow I_Name$ hold
- 3NF Normalization
 - Instructor (I_NIK, I_Name)
 - CourseTeaching (C#, C_Name, I_NIK)

EXAMPLE

- CourseTaking (S_NIM, S_Name, C#, C_Name, I_NIK, I_Name, Grade)



- Third Normal Form
 - Student (S_NIM, S_Name)
 - CourseTaking (S_NIM, C#, Grade)
 - Instructor (I_NIK, I_Name)
 - CourseTeaching (C#, C_Name, I_NIK)

REFERENCES

- Elmasri, Ramez and Shamkant B. Navathe (2016), Fundamentals of Database Systems, 7th edition, Addison Wesley.
- Rick F. van der Lans (2007), SQL for MySQL Developers, Addison Wesley.
- Connolly, Thomas and Carolyn Begg (2015), Database Systems A Practical Approach to Design, Implementation, and Management, 6th edition, Pearson Education, ISBN 13: 978-1-292-06118-4, Essex, England.

```
SELECT *  
FROM WORLD  
WHERE SOMEONE  
LIKE '%YOU%';
```

```
EMPTY SET (0.00 SEC)
```

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2. To conduct research in Informatics to promote science and technology
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